

Youssef Mansour

Alexandria, Egypt | +20 1090110716 | yousseff.mansourrr@gmail.com
github.com/joemans | linkedin.com/in/youssef-mansour-792131246

Full-Stack Developer | Agentic AI & RAG Systems | TypeScript, React, NestJS, Node.js

EDUCATION

Alexandria University — B.Sc. in Computer and Communications Engineering
CGPA: 3.98 / 4.00 Ranked 1st out of 197

Sep 2021 – Jul 2026

EXPERIENCE

Research Intern, Knights Lab

Jul 2025 – Dec 2025

- Reviewed supervised learning and state-space model approaches for video super-resolution, comparing candidate baselines for spatio-temporal reconstruction in a comprehensive survey.
- Set up baseline replication for a Mamba-based video super-resolution architecture, following paper specifications where source code was unavailable.
- Replicated a competitive baseline that approached SOTA performance while using 20% fewer parameters and 34% fewer FLOPs than the compared SOTA model.

PROJECTS

PragmaTrack: AI Prediction Market Tracker

Python, FastAPI, React, LangChain, PostgreSQL, Docker

- Built a full-stack platform aggregating odds from prediction-market APIs such as Kalshi, Polymarket, and Manifold into a unified PostgreSQL schema.
- Developed backend services for API ingestion, normalization, rate-limited endpoints, and structured access to market movement data.
- Integrated a LangChain-based RAG assistant for querying structured market data and generating grounded explanations from retrieved context.
- Exposed the system through a React frontend with interactive visualizations and user-facing analysis workflows.

BuzzShot | [Live](#) | [GitHub](#)

Next.js, React, TypeScript, NestJS, PostgreSQL, Prisma, Redis, Docker, TMDB API

- Built and deployed a full-stack TypeScript monorepo for movie and series discovery with ratings, reviews, watchlists, custom lists, follows, activity feeds, notifications, and personalized recommendations.
- Developed a NestJS API with Prisma/PostgreSQL persistence, Redis caching, JWT access tokens, rotating HttpOnly refresh cookies, Google OAuth, CORS, Helmet, and Zod-based environment validation.
- Implemented a Next.js App Router frontend integrated with TMDB-backed media browsing, search, rich detail pages, authenticated user flows, and reusable shared TypeScript schemas.
- Containerized the local stack with Docker Compose and configured production deployment using Vercel, managed PostgreSQL, Redis, and Prisma migrations.

Car Rental Web Application | [GitHub](#)

React, Node.js, Express, SQL

- Designed and implemented a full-stack car rental platform supporting user authentication, car browsing, reservations, and admin management.
- Built REST API endpoints with Express and SQL-backed persistence for users, vehicles, bookings, and administrative operations.
- Developed a responsive React interface for browsing available cars, managing reservations, and handling core user workflows.

SKILLS

Languages	Python, TypeScript, JavaScript, SQL, C++, Java, Go, Bash
Frontend	React, Next.js, Redux, Zustand, HTML, CSS, Tailwind CSS, Responsive UI
Backend	Node.js, NestJS, Express, FastAPI, Django, REST APIs, JWT Auth, Redis
Databases	PostgreSQL, MySQL, Prisma, SQL Schema Design, Query Optimization
AI/ML	PyTorch, Scikit-Learn, Hugging Face, LangChain, RAG, LLM Fine-Tuning
Testing/Tools	Git, GitHub, Linux, Docker, Vitest, Playwright, Pytest, ESLint, CI/CD
Concepts	Data Structures & Algorithms, OOP, Design Patterns, Software Architecture
Spoken Languages	Arabic (Native), English (IELTS Band 8), German (C1 Goethe Certificate)